# SE LAB-05

## Lab 5: Static Code Analysis

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**Section:** A

**GitHub Link Before Changes (Assigned Problem)**
GitHub - SETAPESU26/L5_60_hostel-room-allocation · GitHub

**GitHub Link After Changes**
**github.com/adityabadde/L5_60_hostel-room-allocation.git**

## Project Summary

This lab applies static code analysis to a Python-based hostel room allocation program. The code was reviewed using Pylint, Bandit, and Flake8 to identify code-quality problems, potential security vulnerabilities, and style or formatting issues.

The identified issues were documented and addressed in the updated version. The work focused on safer configuration and input handling, correcting allocation logic, improving validation, and making the code easier to read and maintain.

## Tools and Results

| Tool | Before Changes | After Changes |
|---|---|---|
| Pylint | 5.84/10 | 10.00/10 |
| Bandit | 5 security findings | No issues identified |
| Flake8 | Formatting/lint issues | No errors reported |

The updated program was also compiled and run successfully as part of the verification workflow.